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Plastic-Waste Composite Concrete — Mix Design Report

Generated	21 October 2026, 20:41 IST
Session ID	TEST-001
Mix Type	PET Aggregate Concrete
Replacement Level	15.0% vol.
Standards Scope	IS 456 / IS 1237 / IS 2185 (BIS, India)
Model Version	PlastiCrete AI v1.0 — M1/M2/M3/M4 Ensemble

Mix Recipe Proportions (per m³)

Design basis: IS 10262 — M20 reference mix adjusted for **15% PET plastic aggregate** replacement, cured at 27.0°C for 28 days.

Constituent	Quantity (kg/m³)	Standard / Grade	Notes
OPC 53 Cement	344.7	IS 12269	Binder, 90% of total
Fly Ash (Class F, IS 3812)	38.3	IS 3812 / IS 15388	10.0% by mass of binder
Fine Aggregate (River Sand)	578.0	IS 383 Zone II	Partially replaced by plastic
Coarse Aggregate (20mm)	1028.5	IS 383	Crushed granite
Polyethylene Terephthalate (PET)	94.5	Recycled, washed	15.0% vol. replacement, 4.0mm
Potable Water	172.3	IS 456 Cl. 5.4	w/c = 0.45

AI-Predicted Mechanical & Physical Properties

Ensemble of XGBoost, Random Forest, and PyTorch DNN trained on 2,000+ peer-reviewed data points. 90% confidence intervals are derived from bootstrap variance across ensemble members.

Property	Predicted Mean	90% CI Low	90% CI High	Unit
Compressive Strength	32.10	29.50	35.20	MPa
Flexural Strength	4.80	4.20	5.40	MPa
Split Tensile Strength	2.90	2.50	3.30	MPa
Bulk Density	2212.00	2170.00	2258.00	kg/m³
Water Absorption	4.10	3.50	4.80	%
Thermal Conductivity	1.18	1.04	1.32	W/m·K
Durability Index	74.00	68.00	80.00	0–100

✓ **Prediction confidence is within normal operating range of the ensemble models.**

BIS Standards Compliance Audit

Standard	Threshold Criteria	Status
IS 1237:2012 — Floor/Paving Tiles	CS ≥ 30 MPa, WA ≤ 1%	**PASS**

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IS 2185:2005 Part 1 — Hollow Blocks	CS ≥ 3.5 MPa	PASS
IS 456:2000 — Plain & Reinforced Concrete	M20 (CS ≥ 20 MPa)	PASS
IS 5816:1999 — Split Tensile Ratio	fst/fck ≥ 0.10	PASS

✓ Mix passes overall BIS compliance threshold.

Life Cycle Assessment (LCA) & Sustainability Score

Cradle-to-gate embodied carbon accounting using ICE Database v3.0 emission factors (Hammond & Jones, 2019). Baseline: conventional IS 10262 M20 mix at 365.7 kg CO■e/m³.

LCA Metric	Value
Embodied Carbon	278.4 kg CO■e/m³
CO■ Saving vs Conventional M20	–23.8%
Plastic Waste Diverted from Landfill	168.0 kg/m³
Equivalent PET Bottles Diverted	~3,696 bottles/m³

Green Building Rating Credits

Green Rating Scheme	Credits Earned	Criterion
IGBC Green Homes / Buildings	3.0 pts	Recycled Content + Thermal
GRIHA (BEE/TERI)	2.0 pts	Embodied Energy, Waste
LEED v4 Materials & Resources	2.0 pts	Recycled Content %
Total Green Credits	7.0 pts	All Schemes Combined

Overall Sustainability Score: 76/100 (Grade A)

Recommended Application & Cost–Benefit Analysis

Primary recommended application: Paving Block Heavy (RF classifier confidence: 89.2%). Product yield: 7.0 m²/m³.

Cost Metric	Plastic-Waste Concrete	Conventional Concrete	Saving
Cost per m³ (INR)	■3,950	■4,850	■900 (18.6%)
Cost per m² product	■564	■692	■128
Plastic aggregate input	168 kg/m³	0 kg/m³	—

Feature Influence Analysis (SHAP — Compressive Strength)

SHAP (SHapley Additive exPlanations) values quantify each input feature's marginal contribution to the predicted compressive strength. Positive = increases CS; Negative = decreases CS vs model baseline.

Feature	SHAP Contribution (MPa)	Direction
replacement_pct=15%	-3.80	▼ Negative
curing_days=28d	+3.50	▲ Positive
additive=fly_ash	+3.10	▲ Positive

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wc_ratio=0.45	+2.40	▲ Positive
particle_size=4.0mm	-0.70	▼ Negative

Remediation Advisory

Top limiting factor: Aggregate replacement strength loss. Implementing the actions below is estimated to improve the sustainability score by **+8.5 points**.

1. Lower w/c ratio to 0.42 using PCE superplasticiser to recover ~3.5 MPa.

2. Increase silica fume to 12% to densify ITZ.

3. Wash plastic aggregates with silane coupling agent.

Literature Analogues — Vector RAG Retrieval (Top 3)

Nearest peer-reviewed mixes retrieved by FAISS L2 search across a 15-dimensional property embedding space (15D = 7 mechanical + 8 mix parameters).

Rank	Authors (Year)	Journal	Application	CS (MPa)	DOI
1	Alkharisi and Dahish (2025)	Construction and Building Mate...	Paving Block Light	24.2	10.1016/j.conbuildmat.2025.1

About PlastiCrete.ai: An explainable ML platform for circular-economy concrete design, predicting 7 mechanical properties via XGBoost + Random Forest + DNN ensemble trained on 2,000+ rows of peer-reviewed plastic-waste concrete research data. Developed for academic, industry, and policy research under Indian Standard engineering frameworks.

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